Accelerating anisotropic elastic subsurface imaging with dual-pair finite differences

Modeste Irakarama, Simon Luo, John Etgen, Xukai Shen, and Michael O'Brien, bp*

SUMMARY

Anisotropic elastic full waveform inversion based on finite difference modeling on fully staggered grids like the Lebedev grid comes with a high computational cost. We examine using a non-staggered finite difference method instead, as a possibility to accelerate elastic subsurface imaging based on fully staggered grids. Most existing studies on finite difference elastic modeling on non-staggered grids are limited to modeling waves sampled above 4 points per P wavelength. This makes them relatively inefficient for subsurface imaging using P waves, compared to the Lebedev grid which can accurately model waves sampled below 3 points per P wavelength for a suitable stencil width. We show that accurate finite difference wave modeling on non-staggered grids below 3 points per P wavelength can be achieved by a combination of dual pairing, selective filtering, and some adjustments to source injection. Preliminary tests on field data show that the proposed methodology can lead to roughly 3 times faster subsurface elastic imaging using P waves, compared to using the Lebedev grid for the same sampling rate, even below 3 points per P wavelengths.

INTRODUCTION

Elastic full waveform inversion (EFWI) has proven to be more reliable in imaging complex subsurface structures compared to alternatives (Elebiju et al., 2022). Elastic subsurface imaging, however, comes with a high computational cost. In this paper, we evaluate the possibility of accelerating finite difference (FD) based tilted transversely isotropic (TTI) EFWI by using more efficient propagators. A TTI EFWI workflow typically relies on the rotated staggered grid (Saenger and Bohlen, 2004) or the Lebedev grid (Lisitsa and Vishnevskiy, 2010) FD modeling for forward and adjoint wave propagation. The Lebedev grid, which is arguably more efficient than the rotated staggered grid, decouples into four standard staggered grids (Bernth and Chapman, 2011). It is therefore reasonable to wonder whether there exist FD implementations of TTI elastic modeling on a non-staggered grid that can accelerate EFWI based on the Lebedev grid up to a factor of four.

Several FD implementations of TTI elastic modeling on non-staggered grids have been presented in the literature (Petersson and Sjögreen, 2015; Dovgilovich and Sofronov, 2015; Duru et al., 2022) but they compare unfavorably to the Lebedev grid, at least for the purpose of elastic subsurface imaging using P waves. Their biggest computational disadvantage is that, in practice, they are mainly suitable for modeling waves sampled above 4 points per P wavelength (PPWL). This is inefficient compared to Lebedev FD which accurately models waves sampled down to $\sim$ 2.7 PPWL using a 17-point wide FD stencil. For computational efficiency, we are primarily interested in wave modeling in the range 4.0-2.5 PPWL; we are not aware of any successful presentation of FD modeling of elastic waves with TTI anisotropy on a non-staggered grid within this sampling range. We show that accurate FD TTI modeling of waves on a non-staggered grid in the PPWL range of interest can be achieved by an appropriate combination of selective filtering (Tam et al., 1993) and dual pairing. Dual pairing, in this paper, refers to the consecutive application of backward and forward finite difference first derivative operators to produce a second derivative operator, with or without a variable coefficient in between. The idea can be found in Liu (1996) and has previously been applied to anisotropic modeling of waves within the summation-by-parts framework (Dovgilovich and Sofronov, 2015; Duru et al., 2022). Here, we relax the summation-by-parts property; we use the image method (Robertsson, 1996) to impose free surface boundary conditions and the two-step implementation of convolutional perfectly matched layers (Fang et al., 2022) to impose absorbing boundary conditions. We present least-squares optimized dual-pair FD operators for 9, 13, 15, 17-point wide stencils designed for modeling waves sampled down to 4.0, 3.1, 2.9, 2.7 PPWL respectively; for each dual-pair FD operator, we present a corresponding selective filter. We show that dual-pair FD can accelerate EFWI imaging up to $\sim$ 3 times compared to Lebedev FD.

THEORY

Given a function $f(x)$ and a variable coefficient $b(x)$, the dual-pair approximation of the second derivative is given by

$$\partial_x\left[b(x_j)\partial_x f(x_j)\right] \approx D^-\left[b(x_j)D^+ f(x_j)\right],$$

where the n-point wide forward operator has the form

$$D^+ f(x_j) = \sum_{m=-nl}^{nr} c_m f(x_j + dx \cdot m),$$

with

$$nl = \frac{n-1}{2} - 1, \quad nr = \frac{n-1}{2} + 1,$$

for some coefficients c_m (see example in Table 1). The backward operator D^- is minus the transpose of D^+ (Dovgilovich and Sofronov, 2015). It has recently been observed that dual-pair FD eliminates the need for artificial damping when solving the TTI elastic wave equation on non-staggered grids (Duru et al., 2022; Dovgilovich and Sofronov, 2015), compared to centered FD (Petersson and Sjögreen, 2015). This appears to be true only for modeling waves sampled above 5 PPWL, as illustrated in Figure 1. The figure shows an experiment in a two-layered synthetic model, with $Vp = 2000$ m/s in the top layer and $Vp = 2500$ m/s in the bottom layer. The model is discretized with cell sizes of $dx = dy = dz = 9$ m, $Vs = 0.57 * Vp$,

10.1190/image2025-4313380.1

and all the simulations were performed using 9-point wide FD operators. Waves computed using a Ricker wavelet of 17 Hz dominant frequency ($\sim$ 38 Hz maximum frequency, $\sim$ 5.8 PPWL) by dual-pair FD and Lebedev FD are shown in Figures 1a and 1b respectively. The two wavefields are consistent. However, increasing the dominant frequency beyond 17 Hz introduces significant numerical artifacts in wavefields generated by dual-pair FD. As an example, Figure 1c shows that Lebedev FD accurately models waves using a dominant frequency of 25Hz ($\sim$ 56 Hz maximum frequency, $\sim$ 3.9 PPWL), whereas dual-pair struggles as shown in Figure 1d. The FD operator used in Figures 1a and 1d is given in the first row of Table 1, and is based on Taylor series. The artifacts in Figure 1d occur at source and material discontinuities and increasing the width of the FD operator, or optimizing the operators to minimize dispersion/dissipation errors, does not improve the results: this is seen in Figure 1e which uses the optimized operator in the second row of Table 1, obtained using a least-squares approach similar to Liu (2014).

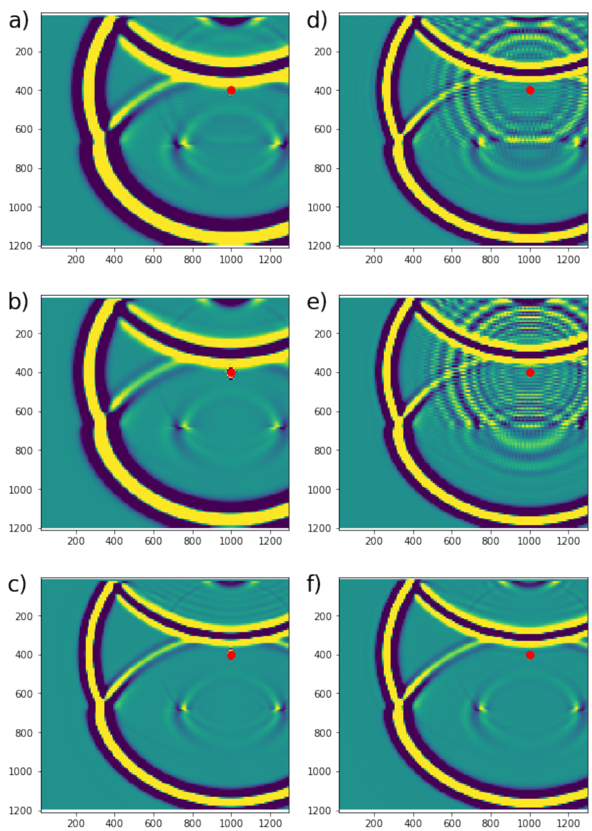

Figure 1: Limitation of dual pairing and proposed solution. a) Dual-pair FD modeling at 5.8 PPWL. b) Lebedev FD modelling at 5.8 PPWL. c) Lebedev FD modelling at 3.9 PPWL. d) Dual-pair FD modeling at 3.9 PPWL. e) Dual-pair FD modeling at 3.9 PPWL using optimized coefficients. f) Dual-pair FD modeling at 3.9 PPWL using optimized coefficients and selective damping. Red dot is the source location.

Dispersion and dissipation errors of the two operators in Table 1 are shown in Figure 2; this figure suggests that the optimized operator should be able to accurately model waves sampled down to $\sim$ 4.0 PPWL since only dispersion/dissipation errors below 0.1% are introduced at that sampling rate.

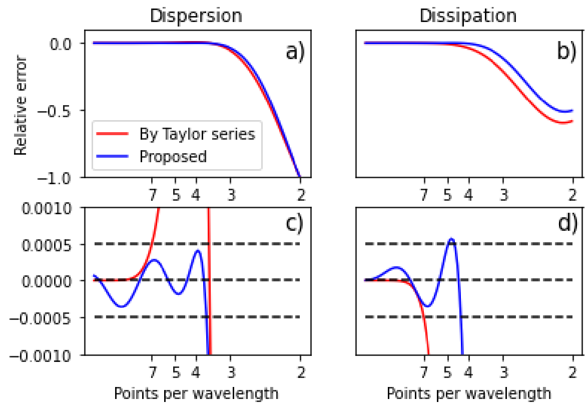

Figure 2: Comparing errors between our FD operator and Taylor series based. a) Dispersion errors. b) Dissipation errors. c) Zoom on dispersion errors within 0.1%. d) Zoom on dissipation errors within 0.1%. Top/bottom dashed line represent zone within 0.05% error.

The numerical artifacts in Figure 1e can be eliminated by selective filtering (Tam et al., 1993), as seen in Figure 1f which is obtained after applying damping using the filter in the third row of Table 2. The filter in the first row is based on Taylor series and the filter in the second row is from Bogey and Bailly (2004). Damping responses of the filters in the table are shown in Figure 3. Our filters are similar to Bogey and Bailly (2004)'s with differences being merely cosmetic: ours are designed to have a maximum of low wavenumbers within 0.01% of ideal response (within dashed lines in 3b) while the filters in Bogey and Bailly (2004) are designed to minimize variations in the damping profile. From a practical standpoint, the two filters are equivalent and numerical experience confirms they both eliminate numerical artifacts without harming waves sampled down to $\sim$ 4.0 PPWL. We have also developed optimized dual-pair operators for 13, 15, 17-point wide stencils, along with a suitable selective filter for each stencil, designed for modeling waves sampled down to $\sim$ 3.1, 2.9, 2.7 respectively. For modeling below 3.0 PPWL using 15 or 17-point stencils, selective filtering alone does not eliminate all artifacts generated at source discontinuity, we include a smoothing constraint in source injection to eliminate remaining artifacts. For examples presented here, we use a 0-th order smoothing constraint in terminology of Petersson et al. (2016).

10.1190/image2025-4313380.1

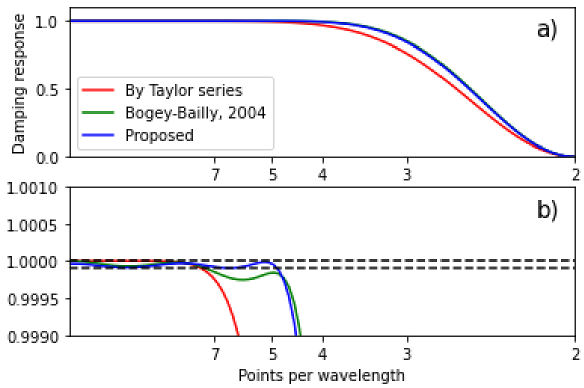

Figure 3: Comparing damping response between our selective filter and others. a) Damping response. b) Zoom on damping response within 0.1% of ideal response. Bottom dashed line represents zone within 0.01% of ideal response.

ELASTIC IMAGING & PERFORMANCE ANALYSIS

We now test dual pairing, on field data, for EFWI imaging using P waves sampled at ~ 2.6 PPWL. We use the initial P velocity model shown in Figure 4a; the model is of size $236 \times 232 \times 101$ with $dx = dy = dz = 100\,m$. Figure 4b is a clipped vertical derivative of the P velocity model, such an image will be referred to as an FDR (FWI Derived Reflectivity) image hereafter. The initial model is used for 5 Hz (dominant frequency) EFWI, then a higher resolution model of size $357 \times 350 \times 152$ with $dx = dy = dz = 66\,m$ is used for 8 Hz EFWI, then a higher resolution model of size $471 \times 462 \times 201$ with $dx = dy = dz = 50\,m$ is used for 11 Hz EFWI. Figures 4c-4f compare Lebedev and dual-pair FDR images obtained after 20 iterations of 11 Hz EFWI.

The following performance numbers are based on the $357 \times 350 \times 152$ model (8 Hz EFWI); while the numbers are, strictly speaking, only valid for our current implementations of Lebedev and dual-pair propagators, we expect them to give a fair estimate of the computational advantage of dual pairing for TTI elastic modeling in general. An elastic Lebedev FD propagator has two main computationally intensive elastic kernels: the stress update kernel, and the displacement update kernel. These two elastic kernels have roughly the same computational cost. Each elastic kernel in Lebedev FD is ~ 3.3 times more expensive than the corresponding kernel in dual-pair FD. However, dual-pair FD has an additional computational kernel: the selective damping kernel. The selective damping kernel has the cost of 1/4 of a single dual-pair elastic kernel; this brings the speed up down to

$$\frac{3.3+3.3}{1+1+1/4} \approx 2.9.$$

IO cost is 1/3 of a single dual-pair elastic kernel, bringing the final speed up to

$$\frac{3.3+3.3+1/3}{1+1+1/4+1/3} \approx 2.7.$$

In terms of memory consumption, dual-pair FD uses roughly 20% less memory than Lebedev FD, which provides an additional advantage for larger models in higher frequency EFWI.

In practice, we generally observe a speed up between ~ 2.0 and ~ 3.2 depending on the size of the model, with higher speed up for larger models. As an example, for the results in Figure 4, dual-pair EFWI was faster compared to Lebedev by a factor of 2.5, 2.7, and 3.2 for 5 Hz, 8 Hz, and 11 Hz EFWI respectively.

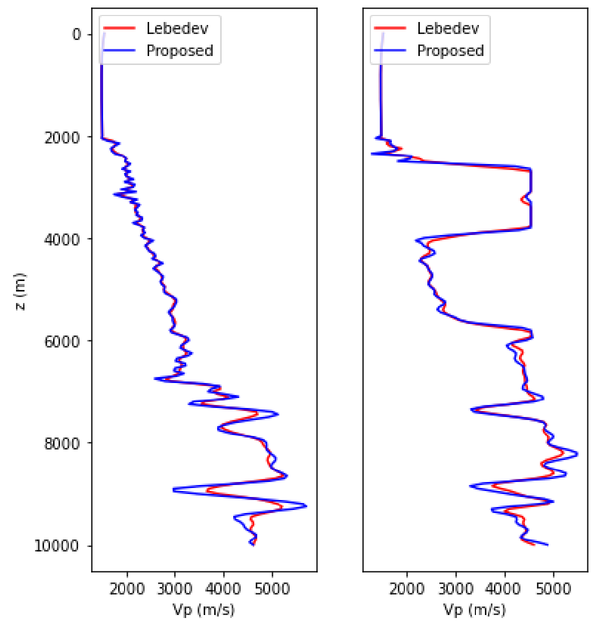

Figure 5: Velocity profiles along the black lines in Figures 4e,4f after 20 iterations of 11 Hz EFWI.

CONCLUSION

We present a methodology for modeling TTI elastic waves using finite differences on non-staggered grids. Unlike previous related efforts, we present successful modeling of waves sampled below 3 PPWL. This is achieved by a combination of dual pairing, selective filtering, and an adjustment to source injection. For this purpose, we develop optimized dual-pair FD operators up to a 17-point wide stencil, and a suitable selective filter for each of the FD operators. We test and apply our method to EFWI subsurface imaging using P waves. We show that the proposed method, while being 3 times faster, can generate comparable results to the ones using Lebedev FD. The EFWI results presented here are preliminary and additional research is ongoing to resolve the differences between the two images observable in the deeper section (e.g. in Figure 5).

ACKNOWLEDGMENTS

We thank bp for permission to publish this work and bp HPC team for support. Seismic data courtesy of VIRIDIEN.

	$j-3$	$j-2$	$j-1$	j	$j+1$	$j+2$	$j+3$	$j+4$	$j+5$
By Taylor series	-0.0059524	0.0714286	-0.5	-0.45	1.25	-0.5	0.1666667	-0.0357143	0.0035714
Proposed	-0.0139592	0.1121395	-0.5906789	-0.3410611	1.1866666	-0.5047765	0.2006772	-0.057683	0.0086754

Table 1: First derivative forward operator 9-point FD stencils used for dual-pair FD.

	j	$j+1$	$j+2$	$j+3$	$j+4$	$j+5$
By Taylor series	0.2460937	-0.2050781	0.1171875	-0.0439453	0.0097656	-0.0009766
Bogey-Bailly, 2004	0.2150449	-0.1877729	0.1237559	-0.0592276	0.0187216	-0.0029995
Proposed	0.2178434	-0.1894915	0.1233971	-0.0577743	0.0176902	-0.0027252

Table 2: Selective filter coefficients for a 9-point FD stencil. The filters are symmetric, only half the stencil is shown.

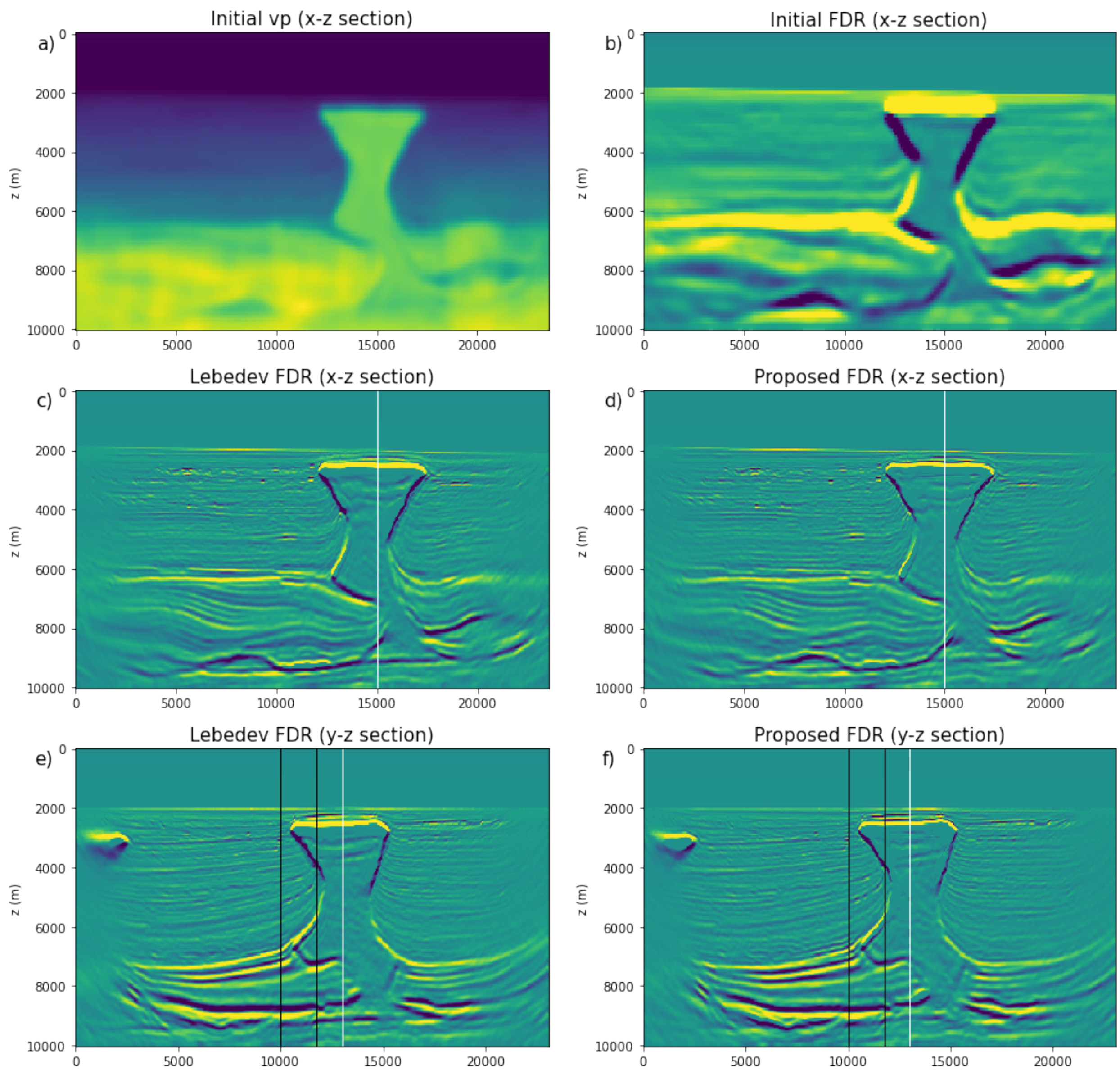

Figure 4: Application of dual-pair FD to EFWI imaging at 2.6 PPWL. a) Initial Vp velocity. b) Initial FDR image. c) x-z section of 11 Hz Lebedev FDR image. d) x-z section of 11 Hz Dual-pair FDR image. e) y-z section of 11 Hz Lebedev FDR image. f) y-z section of 11 Hz Dual-pair FDR image. White lines represent intersection of the two cross sections, black lines represent the location of velocity profiles shown in Figure 5.

10.1190/image2025-4313380.1

REFERENCES

Bernth, H., and C. Chapman, 2011, A comparison of the dispersion relations for anisotropic elastodynamic finite-difference grids: Geophysics, **76**, WA43–WA50, doi: https://doi.org/10.1190/1.3555530.

Bogey, C., and C. Bailly, 2004, A family of low dispersive and low dissipative explicit schemes for flow and noise computations: Journal of Computational Physics, **194**, 194–214, doi: https://doi.org/10.1016/j.jcp.2003.09.003.

Dovgilovich, L., and I. Sofronov, 2015, High-accuracy finite-difference schemes for solving elastodynamic problems in curvilinear coordinates within multiblock approach: Applied Numerical Mathematics, **93**, 176–194, doi: https://doi.org/10.1016/j.apnum.2014.06.005.

Duru, K., F. Fung, and C. Williams, 2022, Dual-pairing summation by parts finite difference methods for large scale elastic wave simulations in 3d complex geometries: Journal of Computational Physics, **454**, 110966, doi: https://doi.org/10.1016/j.jcp.2022.110966.

Elebiju, B., Q. Li, K. Hartman, F. Rollins, Y. Feng, K. Kaiser, W. Schinagl, C. Chen, A. Y. Hao, Z. Wei, and J. Mei, 2022, One more stride forward in thunder horse subsalt imaging with elastic FWI: Second International Meeting for Applied Geoscience & Energy, SEG and AAPG, 947–951, doi: https://doi.org/10.1190/image2022-3750987.1.

Fang, X., D. Wu, F. Niu, and G. Yao, 2022, A new implementation of convolutional pml for second-order elastic wave equation: Exploration Geophysics, **53**, 501–516, doi: https://doi.org/10.1080/08123985.2021.1993441.

Lisitsa, V., and D. Vishnevskiy, 2010, Lebedev scheme for the numerical simulation of wave propagation in 3d anisotropic elasticity: Geophysical Prospecting, **58**, 619–635, doi: https://doi.org/10.1111/j.1365-2478.2009.00862.x.

Liu, Y., 1996, Fourier analysis of numerical algorithms for the maxwell equations: Journal of Computational Physics, **124**, 396–416, doi: https://doi.org/10.1006/jcph.1996.0068.

Liu, Y., 2014, Optimal staggered-grid finite-difference schemes based on least-squares for wave equation modelling: Geophysical Journal International, **197**, 1033–1047, doi: https://doi.org/10.1093/gjihttps://doi.org//ggu032.

Petersson, N. A., O. O'Reilly, B. Sjogreen, and S. Bydlon, 2016, Discretizing singular point sources in hyperbolic wave propagation problems: Journal of Computational Physics, **321**, 532–555, doi: https://doi.org/10.1016/j.jcp.2016.05.060.

Petersson, N. A., and B. Sjogreen, 2015, Wave propagation in anisotropic elastic materials and curvilinear coordinates using a summation-by-parts finite difference method: Journal of Computational Physics, **299**, 820–841, doi: https://doi.org/10.1016/j.jcp.2015.07.023

Robertsson, J. O. A., 1996, A numerical free-surface condition for elastic/viscoelastic finite-difference modeling in the presence of topography: Geophysics, **61**, 1921–1934, doi: https://doi.org/10.1190/1.1444107.

Saenger, E. H., and T. Bohlen, 2004, Finite-difference modeling of viscoelastic and anisotropic wave propagation using the rotated staggered grid: Geophysics, **69**, 583–591, doi: https://doi.org/10.1190/1.1707078.

Tam, C. K., J. C. Webb, and Z. Dong, 1993, A study of the short wave components in computational acoustics: Journal of Computational Acoustics, **1**, 1–30, doi: https://doi.org/10.1142/S0218396X93000020.